

Week 2 - Assignment 2: ML Task Framing Guide

Objective: Frame your chosen lane as a Machine Learning problem before building any model.

Assignment Requirements

Complete the notebook `work/notebooks/w02_ml_task_framing.ipynb`. Fill in: 1. My Lane as an ML Task 2. Target or Proxy 3. Success Metric 4. Unit of Analysis (show a real dataframe) 5. Why Machine Learning Beats a Fixed Rule 6. Self-check

Recommended Answers

Selected Lane: Content Refresh Prioritization

ML Task Type: Ranking

Target / Proxy: Refresh Priority Score for each webpage.

Success Metric: Precision@K or NDCG.

Unit of Analysis: One row represents one webpage containing search metrics such as clicks, impressions, CTR, and average position.

Why Machine Learning? A simple rule such as 'refresh every page with low CTR' ignores many interacting signals. Machine Learning combines multiple features to prioritize pages more accurately than a fixed handcrafted rule.

Notebook Code Example

Load the starter dataset and display the first rows:

```
df = pd.read_csv('../data/raw/content_refresh_anonymized.csv')
df.head()
```

Submission

Run every notebook cell, save the notebook, commit it to GitHub, and submit your repository URL on the FlyRank assignment page.